

Breast Cancer Wisconsin Diagnostic

Machine Learning Project Report & Comprehensive Pipeline Analysis

1. Problem Statement

Breast cancer diagnosis is a binary classification problem in which a tumor is classified as either **Benign (B)** or **Malignant (M)**.

The objective of this project is to develop a machine learning system that can classify breast tumors using quantitative measurements extracted from cell nuclei. The project uses the **Breast Cancer Wisconsin Diagnostic (WDBC)** dataset, which contains 569 samples and 30 numerical features describing characteristics such as radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension.

Dataset Breakdown & Preprocessing Summary:

- **Total Samples:** 569
- **Input Features:** 30 numerical attributes
- **Benign Samples:** 357 (62.7%) — Encoded as 0
- **Malignant Samples:** 212 (37.3%) — Encoded as 1
- **Removed Columns:** id (identifier) and Unnamed: 32 (completely empty)

2. Project Architecture

The project follows an end-to-end machine learning pipeline:

Raw Dataset → Data Inspection → Data Cleaning → EDA → Feature Scaling → PCA → Unsupervised Clustering → Train-Test Split → Supervised Models → Evaluation → Hyperparameter Tuning → Model Selection & Export

3. Data Preprocessing

Two columns were removed because they were not useful for prediction: id and Unnamed: 32. The diagnosis target column was encoded using LabelEncoder (B → 0, M → 1). After cleaning, zero missing values remained across all 569 rows and 31 columns.

4. Exploratory Data Analysis (EDA)

Correlation analysis was performed between numerical features and tumor diagnosis target. The features displaying the highest absolute correlation with diagnosis include:

Feature Name	Absolute Correlation
concave points_worst	0.7936
perimeter_worst	0.7829
concave points_mean	0.7766

Feature Name	Absolute Correlation
radius_worst	0.7765
perimeter_mean	0.7426
area_worst	0.7338
radius_mean	0.7300
area_mean	0.7090
concavity_mean	0.6964
concavity_worst	0.6596

5. Feature Scaling and Dimensionality Reduction

Because feature scales varied widely, `StandardScaler` was applied to normalize inputs. Principal Component Analysis (PCA) was subsequently evaluated for feature simplification:

- **Original Features:** 30
- **Components required for 95% Variance:** 10 components
- **Components required for 99% Variance:** 17 components

6. Unsupervised Learning

Unsupervised clustering was conducted to evaluate whether tumor features naturally separate into benign and malignant groupings without label guidance.

6.1 K-Means Clustering Results (K = 2 to 10)

K Value	Silhouette Score
2	0.3433
3	0.3144
4	0.2835
5	0.1586
6	0.1607
7	0.1538
8	0.1479
9	0.1471
10	0.1370

For **K = 2**, K-Means yielded a Silhouette Score of 0.3434 and an Adjusted Rand Index (ARI) of 0.6536.

Cluster-to-Diagnosis Contingency Matrix (K-Means)

True Diagnosis	Cluster 0	Cluster 1
Benign	339	18
Malignant	36	176

6.2 Agglomerative Hierarchical Clustering

- ARI: 0.5750
- Silhouette Score: 0.3394

7. Supervised Learning & Evaluation

Dataset split: 80/20 Stratified Train-Test split (455 training samples, 114 testing samples).

8 & 9. Initial Model Comparison

Model	Accuracy	F1-Score	ROC-AUC	5-Fold CV F1
Logistic Regression	0.9649	0.9512	0.9960	0.9633
Decision Tree	0.9298	0.9048	0.9246	0.8757
Random Forest	0.9737	0.9630	0.9929	0.9381
Gradient Boosting	0.9649	0.9500	0.9947	0.9315
SVM (RBF)	0.9737	0.9630	0.9954	0.9666
KNN	0.9561	0.9383	0.9816	0.9532
Gaussian Naive Bayes	0.9211	0.8889	0.9891	0.9045
XGBoost	0.9737	0.9630	0.9940	0.9360

10 & 11. Hyperparameter Tuning Results

Hyperparameter tuning via GridSearchCV with 5-fold cross-validation optimized for F1-score produced the following optimal metrics:

Tuned Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
SVM (RBF) ★	0.9825	1.0000	0.9524	0.9756	0.9960
Random Forest	0.9737	1.0000	0.9286	0.9630	0.9929
XGBoost	0.9737	1.0000	0.9286	0.9630	0.9937
Logistic Regression	0.9649	0.9750	0.9286	0.9512	0.9960

12 & 14. Final Recommended Solution

Champion Model: Support Vector Machine (RBF Kernel)

- **Best Hyperparameters:** $C = 10$, $\gamma = 0.01$
- **Test Accuracy:** 98.25%
- **Precision:** 100.00% (Zero false positives in malignant predictions)
- **Recall:** 95.24% (High sensitivity for identifying positive cases)
- **F1-Score:** 97.56%
- **ROC-AUC:** 99.60%

15 & 16. Limitations & Ethical Disclaimer

Ethical Disclaimer:

This project is intended strictly for educational and research purposes. It must not be used as a standalone clinical diagnostic system. Machine learning models should never replace medical evaluation by qualified healthcare professionals or clinical laboratory testing.

17. Project Checklist — 20 Key Completed Steps

- ✓ Define the breast cancer classification problem
- ✓ Load the WDBC dataset
- ✓ Inspect dataset dimensions
- ✓ Inspect data types
- ✓ Check missing values
- ✓ Analyze class distribution
- ✓ Remove irrelevant columns
- ✓ Encode the diagnosis target
- ✓ Perform exploratory data analysis
- ✓ Analyze feature correlations
- ✓ Standardize numerical features
- ✓ Apply PCA for dimensionality reduction
- ✓ Perform K-Means clustering
- ✓ Evaluate clustering using Silhouette Score
- ✓ Evaluate clustering using ARI
- ✓ Perform Agglomerative Clustering
- ✓ Train multiple supervised classifiers
- ✓ Evaluate models using multiple metrics
- ✓ Perform hyperparameter tuning

✓ Select and save the final model

18. Conclusion

This project delivered a comprehensive ML pipeline for breast cancer diagnosis. The tuned **RBF Support Vector Machine (SVM)** achieved exceptional diagnostic accuracy (98.25%), precision (100.00%), and ROC-AUC (99.60%), proving its suitability for automated diagnostic support workflows.